

Falcon 220A alternator diode failure causing system voltage low and battery c

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Area: bulletins

1 Condition

Affected Falcon vehicles may set DTC P0562 (System Voltage Low) accompanied by a discharged battery BAT-1210 and a charge warning indicator. The customer may report no-start conditions after the vehicle sits overnight, or dim headlamps at idle. Confirm the low-voltage complaint with a charging system load test before parts replacement, and review the Charging System Overview for expected output values. The System Voltage Codes section describes the P0562 set criteria in detail.

2 Cause

The condition is caused by a failed rectifier diode inside the 220A alternator ALT-8812, which allows reverse current leakage and produces both the P0562 fault and an excessive parasitic battery drain. The degraded diode reduces charging output below the regulation threshold under electrical load. Refer to the Alternator overview for the ALT-8812 internal layout, and the Battery section for how the drain affects BAT-1210 state of charge.

3 Repair Procedure

Start with PROC-CHG-DIAG to confirm the low charging output and isolate ALT-8812 as the source, following the Charging System Diagnosis Procedure . Replace alternator ALT-8812 per PROC-ALT and verify regulated output returns to specification under load. If battery BAT-1210 fails a post-repair capacity test from the prior deep discharge, replace it using the Battery Replacement Procedure in the Battery Sub-System Service section. Clear P0562, recharge or replace BAT-1210 as needed, and confirm no recurring drain. The Generator Control Circuit Code reference may help rule out related control-circuit faults.